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ABSTRACT

Purpose: To screen the entire genome for genetic markers associated with risk for concussion.

Methods: A genome-wide-association (GWA) analyses was performed utilizing data from the Kaiser Permanente Research Board (KPRB) and the United Kingdom (UK) Biobank. Concussion cases were identified based on electronic health records from KPRB and UK Biobank from individuals of European ancestry. Genome-wide association analyses from both cohorts were tested for concussion using a logistic regression model adjusting for sex, height, weight and race/ethnicity using allele counts for single nucleotide polymorphisms (SNPs). Previously identified genes within the literature were also tested for association with concussion.

Results: There was a total of 4,064 cases of concussion and 291,472 controls within the databases, with two SNPs demonstrating a genome-wide significant association with concussion. The first polymorphism, rs144663795 ($p = 9.7x10^{-11}$; OR=2.91 per allele copy), is located within the intron of *SPATA5*. Strong, deleterious mutations in *SPATA5* cause intellectual disablility, hearing loss and vision loss. The second polymorphism, rs117985931 ($p = 3.97x10^{-9}$; OR= 3.59 per allele copy) is located within *PLXNA4. PLXNA4* plays a key role is axon outgrowth during neural development, and DNA variants in *PLXNA4* are associated with risk for Alzheimer's disease. Previous investigations have identified five candidate genes that may be associated with concussion, but none showed a significant association in the current model ($p < 0.05$).

Conclusion: Two genetic markers were identified as potential risk factors for concussion and deserve further validation and investigation of molecular mechanisms.

Keywords: traumatic brain injury, genetic testing, SPATA5, PLXNA4

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INTRODUCTION

Concussion is a traumatic brain injury (TBI) induced by biomechanical forces (1). It can result from a direct blow to the head or neck area or from an impacted force to another portion of the body with resultant forces to the head. Concussion is common amongst athletes involved in collision sports and soldiers, but can also occur as a result of falls, car accidents, or other events leading to head trauma (2, 3). In the United States, up to 3.8 million sports-related concussions are diagnosed yearly (4) while in England and Wales, 1.4 million people per year seek care at hospitals following a head injury (5). Between 2000 and 2015, there were more than 250,000 military personnel diagnosed with concussion (6).

Sports related concussion manifests itself with a variety of overlapping signs and symptoms such as physical maladies, cognitive impairment, neurobehavioral features, and sleep disturbance. With appropriate treatment of first time concussions, approximately 80-90% of persons experience symptom resolution within 7-10 days (7). Others, particularly those with prior concussions or underlying psychological conditions, can experience prolonged symptoms of months or longer (8). While recovery is expected following a single, uncomplicated concussion, there are many factors that can influence recovery, such as age (9), sex (10-12), history of prior concussions (12, 13), education level (14), premorbid mental health disorders (11, 12, 15) and comorbid mood disorder (16). Repetitive concussions have been linked with long-term neurodegenerative issues including Chronic Traumatic Encephalopathy (CTE)(17, 18).

Prior studies have reported genetic risk factors that are associated with an increased risk of concussion (19-22). These studies have taken a candidate gene approach using genes with

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known neurological functions, with many focusing on the E4 allele of the apolipoprotein E gene (ApoE)(19-25). Other candidate genes that have been tested for association with concussion risk include genes involved in neuronal function (DRD4, SLC17A7, BDNF and NGF), neuronal disease (MAPT) and neuro-inflammation (IL-6/IL-6R (19, 23, 25-29).

In contrast to candidate gene studies that are restricted to testing a limited number of pre-selected genes, genome-wide association (GWA) analyses reveal the strongest associations in the genome by searching in an agnostic fashion without prior bias. In a GWA analysis, millions of polymorphisms throughout the genome are scanned for potential associations with a phenotype such as concussion, in which a variant allele is more prevalent in cases than in controls. To achieve statistical significance, extremely large cohorts of cases and controls are required. Uncovering genetic markers for risk for concussion could help in understanding the biological mechanisms underlying concussion as well as symptoms associated with post-concussion syndrome. Further, genetic markers could be used to identify individuals at risk. This information might lead to improved prevention and treatment for concussion, help inform return-to-play guidelines, and perhaps direct an athlete's choice of sport or training.

In this work, we report the first genome-wide association analysis for concussion using data from the Kaiser Program Research Board (KPRB) and the UK Biobank. We hypothesize that genes without previously documented association with concussion would be identified.

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METHODS

Genome-wide association analyses (GWAS) for concussion were performed using data from the KPRB (with which the Kaiser Permanente, Northern California Research Program in Genes, Environment and Health (RPGEH) is affiliated) and from the v3 release of UK Biobank.

KPRB Cohort

The data generation and data analysis pipelines for the KPRB cohort have been previously described (30). Our analysis cohort includes 83,414 individuals of European ancestry who were genotyped at 670,572 SNPs using Affymetrix Axiom genome-wide arrays. Genotypes were pre-phased with Shape-IT v2.r644 (accessed Feb. 2, 2016) then imputed to a cosmopolitan reference panel consisting of all individuals from the 1000 Genomes Project (Mar 2012 release) using IMPUTE2 v2.2.2 (accessed Feb. 2, 2016) and standard procedures with a cutoff of $R^2 > 0.3$. The final number of SNPs following imputation was 9,878,560. The quality of the imputed data was previously validated (31).

UK Biobank Cohort

Genotype data were obtained from the v3 release of UK Biobank (32). The UK Biobank electronic healthcare records were available for 212,122 individuals and included data until June 2019. Genotype data were imputed centrally by UK Biobank with IMPUTE2 using the Haplotype Reference Consortium and the UK10k+ 1000GP3 reference panels (33). Metrics for quality control were established and then used to filter DNA variants by UK Biobank (32). Imputed SNPs were excluded if they had an IMPUTE2 info score < 0.4.

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Database Quality Control

For both the KPRB and UK Biobank cohorts, individuals were excluded if they were outliers based on genotyping missingness rate or heterogeneity, whose sex inferred from the genotypes did not match their self-reported sex, who withdrew from participation or who were not of European ancestry. The purpose of restricting individuals to those with European ancestry is to reduce population stratification in the study; for example, if the risk of concussion amongst individuals with African ancestry is higher than that for European individuals, then any SNP with an allele frequency that is different between African and European ancestries would appear to be associated with concussion. Overall, these filters resulted in excluding 18.9% and 7.9% of individuals (mostly due to the ancestry filter) in the KPRB and UK Biobank cohorts, respectively. Genetic variants were excluded that failed quality control procedures in any of the genotyping batches, that showed a departure from Hardy-Weinberg of $P<10^{-50}$ or that had a Minor Allele Frequency < .001. Determination of genetic ancestry was performed by principal component analysis (PCA) computed centrally by either KPRB or UK Biobank, as previously described (32).

Phenotype Definitions

In the KPRB cohort, concussion cases were identified based on clinical diagnoses captured in the Kaiser Permanente Northern California electronic health record system prior to July 22, 2015. International Classification of Disease, Ninth Revision (ICD-9) or International Classification of Disease, Tenth Revision (ICD-10) codes were used to identify cases of concussion (Table 1). In the UK Biobank cohort, concussion cases were identified from primary care data, using either Read v2 or Read v3 codes (Table 1).

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Genome-Wide Association

Genome-wide association studies were conducted using PLINK v2.0a (30). SNP associations were tested with concussion with a logistic regression model using allele counts for typed and imputed SNPs. The model was adjusted for genetic sex, height, weight and race/ethnicity using 10 principal components, as previously described (30). For the UK Biobank, the age of injury was available and was included as an adjustment as well, with age of enrollment used for controls. The final number of SNPs that was analyzed was 9,878,560 in the KPRB cohort and 9,856,942 in the UK Biobank cohort. To account for inflation due to population stratification, the genomic control parameter (λgc) was calculated ($\lambda gc = 1.056$ for KPRB; $\lambda gc = 0.925$ for UK Biobank). Subsequently, p-values were adjusted for the genomic control in each population.

Results using odds ratios per allele from each cohort were combined by inverse-variance, fixed-effects meta-analysis as previously described (30). Here, meta-analysis refers to a statistical method to combine data from GWAS's performed on two independent cohorts. The total number of SNPs from either the KPRB or the UK Biobank analysis was 14,721,556; the final number of SNPs that were present in both cohorts, and hence analyzed in the fixed-effects meta-analysis, was 7,743,986. For a p-value threshold, we set a threshold such that the family-wise error rate is 0.05 using the algorithm developed by Kemp et al., 2017 (34); using the total number of SNPs from either cohort (14,721,556 SNPs) yields $p < 7.7E-9$ as a threshold for genome-wide significance. Power calculations were conducted with the software using the Genetic Association Study Power Calculator (accessed Jan. 11, 2020)(35).

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Further bioinformatics investigation of the top genome-wide significant loci from the GWAS was conducted. QQ and Manhattan plots were created using qqman. Regional association plots were generated for each locus with LocusZoom (accessed January 1, 2020)(36). Neither rs144663795 nor rs117985931 were directly genotyped on the Affymetrix chips, but rather their genotype data was imputed. The info scores from IMPUTE2 were 0.92 and 0.89 for rs144663795 and rs117985931, respectively. These info scores indicate that the imputed genotypes are accurate.

The statistical associations for KPRB and UK Biobank were compared to determine if the association with concussion was stronger in one cohort than the other, a phenomenon referred to as heterogeneity (37). rs144663795 did not show strong evidence for heterogeneity between the KPRB and UK Biobank cohorts. For rs117985931, Q and I^2 were 0.0 and 0.89, suggesting that the effects were larger in the UK Biobank than the KPRB cohort.

The genomic context of each SNP was investigated using RegulomeDB (accessed February 1, 2020)(38) web tools. Whether each SNP is an expression quantitative trait locus (eQTL) was queried using the NCBI eQTL Browser (accessed February 1, 2020) and the Genotype-Tissue Expression (GTEx) Portal (accessed February 1, 2020). ChIP seq data from the ENCODE project was used to determine whether SNPs were located within transcription factor binding sites (39). Summary statistics for all SNPs from the GWAS will be available at NIH GRASP: https://grasp.nhlbi.nih.gov/FullResults.aspx upon acceptance of this manuscript.

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Testing of Previously Identified Candidate Genes

A literature search on the genetics of concussion was performed in December, 2019 using "concussion genetics" as search terms in Pubmed. In addition, references from two recent review article on concussion genetics were included (22, 40). Candidate genes were tested for validation when the relevant polymorphisms were present in the summary statistics from the genome-wide analyses performed using the KPRB and UK Biobank data (Table 4).

Ethical Approval

This study analyzed stored data from KPRB and UK Biobank subjects who consented to genomic testing and use of their genomic data, as well as health data from the KPNC and UK Biobank electronic health records. The health and genotype data for the subjects were de-identified. All study procedures were approved by the Institutional Review Board of the Kaiser Foundation Research Institute.

Data sharing

All data will be *openly and publicly* available upon publication of this article.

Patient Involvement

Patients were not involved in this research, except as anonymized subjects in the two cohorts.

RESULTS

Identification of DNA Variants Associated with Concussion

We performed genome-wide association analyses for concussion with the KPRB (83,414 individuals) and UK Biobank (212,122 individuals) cohorts using sex, weight, height and age of

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injury as adjustments (Table 2). For KPRB, there were 3,170 cases of concussion and 80,244 controls (Table 1). For UK Biobank, there were 894 cases and 211,228 controls (Table 1). Manhattan and QQ plots are shown in Supplemental Figure 1 (see Figure, Supplemental Digital Content 1, QQ and Manhattan plots for genome-wide association analysis of concussion, http://links.lww.com/MSS/C166). There were two SNPs with genome-significant associations with concussion using $p=7.7x10^{-9}$ as a cut-off: rs144663795 on chromosome 4 and rs117985931 on chromosome 7 (Table 3, Figure 1 A-B). rs117985931 shows an association with concussion with a p-value ($p = 3.97x10^{-9}$) that is close to the threshold for genome-wide significance, and thus one should be cautious regarding this association until it can be validated in follow-up studies.

Validation of Previous Candidate Gene Studies

The E2, E4 and rs405509 alleles of ApoE have been tested for association with risk of concussion in several previous studies, with mixed results (11-13, 15-17, 45-48). In our analysis, there were a total of 4,064 cases of concussion, and neither the ApoE4 allele ($p = 0.19$) nor rs405509 ($p = 0.31$) was significantly associated with concussion (Table 4). For ApoE2, the association with concussion has a p-value that is borderline significant ($p=.067$) with an odds ratio of 1.11 (95% CI; 0.99 to 1.53).

Previous candidate gene studies have also reported associations with concussion for BDNF, DRD4, SLC17A7 and NGF (19, 26). We tested the polymorphisms in these genes in our concussion dataset, and did not find a significant association for any ($p > 0.05$)(Table 4). In summary, we were not able to replicate the results based on candidate genes from any of the previous studies of concussion.

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DISCUSSION

Genetic Markers for Concussion

This study provides new information about a possible genetic mechanism associated with concussion risk. We demonstrated the first evidence for genetic factors affecting risk for concussion with genome-wide significance. Two SNPs were associated with concussion in the GWAS (rs144663795 and rs117985931). Previously-identified candidate genes within the literature were not confirmed to be associated with concussion in the current dataset.

rs144663795 is in the intron of the Spermatogenesis Associated 5 gene (*SPATA5*), which encodes an ATPase with unknown function. Strong, deleterious missense mutations in *SPATA5* cause severe intellectual disability, hearing loss, vision impairment and slow and asymmetric waves in the EEG (41-43). *SPATA5* function may be weakly affected by rs144663795 with no overt phenotype except for an increased risk of concussion, and pathologic mutations in *SPATA5* may directly cause a strong neurological phenotype.

rs117985931 is located in an intron of *PLXNA4*, which encodes a plexin that is a type of neuronal receptor whose role is to transduce signals from semaphorins to steer axon growth (44). Neural injury causes an increase in expression of *PlexinA4* suggesting a role in neural maintenance and regeneration (45). A genome wide association study revealed that DNA variants in *PLXNA4* are associated with Alzheimer's disease (46, 47). The finding from this study adds an important new insight about the overall roles for plexins, linking their roles in neural development, neural injury, Alzheimer’s disease and concussion.

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Individuals harboring risk alleles for rs144663795 (C) in *SPATA5* or rs117985931 (G) in *PLXNA4* have an increased risk for concussion. The frequency of rs144663795 (C) in *SPATA5* and rs117985931 (G) was 1.4% and 0.13% in the UK Biobank cohort, respectively (Table 3). Although the genetic association results have not yet been validated in an independent study, the two genetic markers could provide key information to athletes and soldiers about their risk for concussion. Genetic testing in either athletes or soldiers could allow them to take extra precautions to avoid concussion, and also compel them to seek clinical treatment that they might otherwise ignore. The genetic information could also be used by medical professionals to make more informed decisions regarding concussion diagnosis, management and return-to-play. Hercher et al. studied whether NCAA athletes would be interested in genetic testing for concussion, and found that 74% expressed interest in undergoing genetic testing for an increased risk of poor recovery from concussion (48).

Validation of Candidate Gene Studies

Previous studies have tested candidate genes for association with concussion, based on the known role of those genes with neurological disorders such as Alzheimer's disease, major depressive disorder and impulsive personality traits. In a study involving college athletes, ApoE4 was found to have a significant association ($p = 0.04$) with the E4 allele acting to protect against concussion (OR = 0.61)(49). In another study involving soldiers, the E4 allele had the opposite effect; it was associated with an increased risk of concussion that was borderline significant ($p=.087$)(19). Other studies found that the E4 allele was associated with either a worse chronic brain injury score ($p = 0.04$)(20) or a more severe total concussion symptom score ($p = 0.015$)(21). Finally, two studies found no significant association of the E4 allele with concussion (23, 24). In contrast to ApoE4, the ApoE2 allele is associated with decreased risk of Alzheimer's

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disease (50, 51). A third allele of ApoE (rs405509) is referred to as the G-219T promoter polymorphism. One study reported a significant association of rs405509 with concussion (p= 0.04)(52), but two other studies with larger cohorts found no significant association (23, 25).

Previous candidate gene studies have reported associations with concussion for BDNF (rs6265) and the CC genotype of DRD4 (rs1800955)(19, 26). Other studies have reported an association between the CC genotype of SLC17A7 (rs74174284) and concussion duration and severity, and the TT genotype of NGF (rs3783988) with better functional outcomes following concussion (27, 28).

Our GWAS contains many more cases of concussions than were available in previous studies (Table 4). For the candidate genes reported in previous studies, we performed follow-up experiments using the GWAS results in an attempt to validate the results using a larger cohort. We were not able to replicate any of the previous reports of association of candidate genes with concussion. Power calculations indicate an 80% chance of success if the genotype relative risk of the candidate gene is at least 1.15. One explanation for the lack of validation is that the previous studies looked at cases of concussion in athletes and soldiers, whereas our study looked at individuals from the general population. Nevertheless, evidence from many other studies suggests that candidate gene associations need to be independently replicated, otherwise their credibility is low (53, 54).

We found that sex, height and weight had opposite effects on risk of concussion in the RPGEH and UK Biobank cohorts. In the RPGEH cohort, being female and weighing less were

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associated with increased risk of concussion, consistent with previous epidemiological studies showing that female athletes are at greater risk for concussion than their male counterparts (55). In the KPRB cohort, however, we found the opposite result; being male, heavier in weight and taller in height resulted in a higher chance of having had a concussion. The association of weight with concussion may be related to the prior observation that obesity in athletes results in increased time to return to baseline after sports-related concussions (56).

There was a higher incidence of concussion cases in the KPRB cohort (3.8%) than in the UK Biobank cohort (0.42%). The electronic records for both cohorts extend for the entire lifetime of the patient if reported by the patient and recorded by the physician. It is unclear whether the difference in concussion frequency reflects a true difference in incidence or whether it is a bias in how concussion is diagnosed in the Bay Area versus the UK.

Limitations

Our analysis found only two genome-wide significant signals, possibly because concussion may be poorly documented in these cohorts. This type of misclassification error would mostly tend to dilute the strength of any signals, if present. Alternatively, it could be that the heritability of concussion phenotypes is low. Another limitation is that the phenotypes were defined from codes contained in electronic health records, and thus we have no information regarding the clinical scenarios surrounding the event. This would include whether patients had prior concussions that were not captured, had a series of sub-concussive events that went unreported, and the force and/or impact velocity of the inciting event. Furthermore, without this clinical information and given the potential inaccuracies of the diagnosis codes within the database, we may be unable to determine whether our identified genetic risk factors place people at increased risk of an initial

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concussion (i.e. lower their threshold for concussion) or rather place them at increased risk of sustaining more severe and prolonged symptoms (i.e. post-concussive syndrome). Additionally, the cohort included people regardless of whether or not they participated in a sport. For example, we were unable to discriminate if the concussions identified in this study were related to participation in sports or from other causes, such as falls, being struck by or against an object, motor vehicle accidents or self harm. Lastly, this study only evaluated individuals from the European ancestry group, and the effect of rs144663795 in *SPATA5* or rs117985931 in *PLXNA4* in other ethnicities is unknown.

CONCLUSION

In the future, it will be important to replicate these gene association results with concussion in independent cohorts, especially for athletes and soldiers. Additional studies are warranted to begin to illuminate the underlying biological mechanism for the association of variation near *SPATA5* or *PLXNA4* with concussion. The results from these studies may validate whether rs144663795 in *SPATA5* or rs117985931 in *PLXNA4* can be used as diagnostic markers to help predict which athletes harbor a higher risk for incidence of concussion or from complications following concussion. Follow-up experiments could look at whether rs144663795 in *SPATA5* or rs117985931 in *PLXNA4* affect other aspects of concussion, such as length of recovery time, response to different types of treatment or risk for chronic traumatic encephalopathy.

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CONFLICT OF INTEREST

SKK is CEO of AxGen, Inc., a genetic testing company for sports injuries. MDR is a consultant for AxGen, Inc. The results of the present study do not constitute endorsement by the American College of Sports Medicine. The results in this work are presented clearly, honestly, and without fabrication, falsification, or inappropriate data manipulation.

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FIGURE LEGEND

Figure 1. A. Regional-association plot for rs144663795 with concussion. Tested SNPs are arranged by genomic position on chromosome 4 (x-axis) in a 600 kb window around the lead SNP rs144663795 (purple diamond). The y-axis indicates -$\log_{10}$ p-values for association with concussion for each SNP. rs144663795 is located in the intron of SPATA5. The color of dots of the flanking SNPs indicates their linkage disequilibrium (R^2) with the lead SNP as indicated by the heat map color key. B. **Regional-association plot for rs117985931 with concussion.** Tested SNPs are arranged by genomic position on chromosome 7 (x-axis) around the lead SNP rs117985931 (purple diamond). The y-axis indicates -$\log_{10}$ p-values for association with concussion for each SNP. rs117985931 is located in an intron of the *PLXNA4* gene. The color of the dots of the flanking SNPs indicates their linkage disequilibrium (R^2) with the lead SNP as indicated by the heat map color key.

Supplemental Digital Content

Supplemental figure—QQ and Manhattan plots for genome-wide association analysis of concussion

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Figure 1

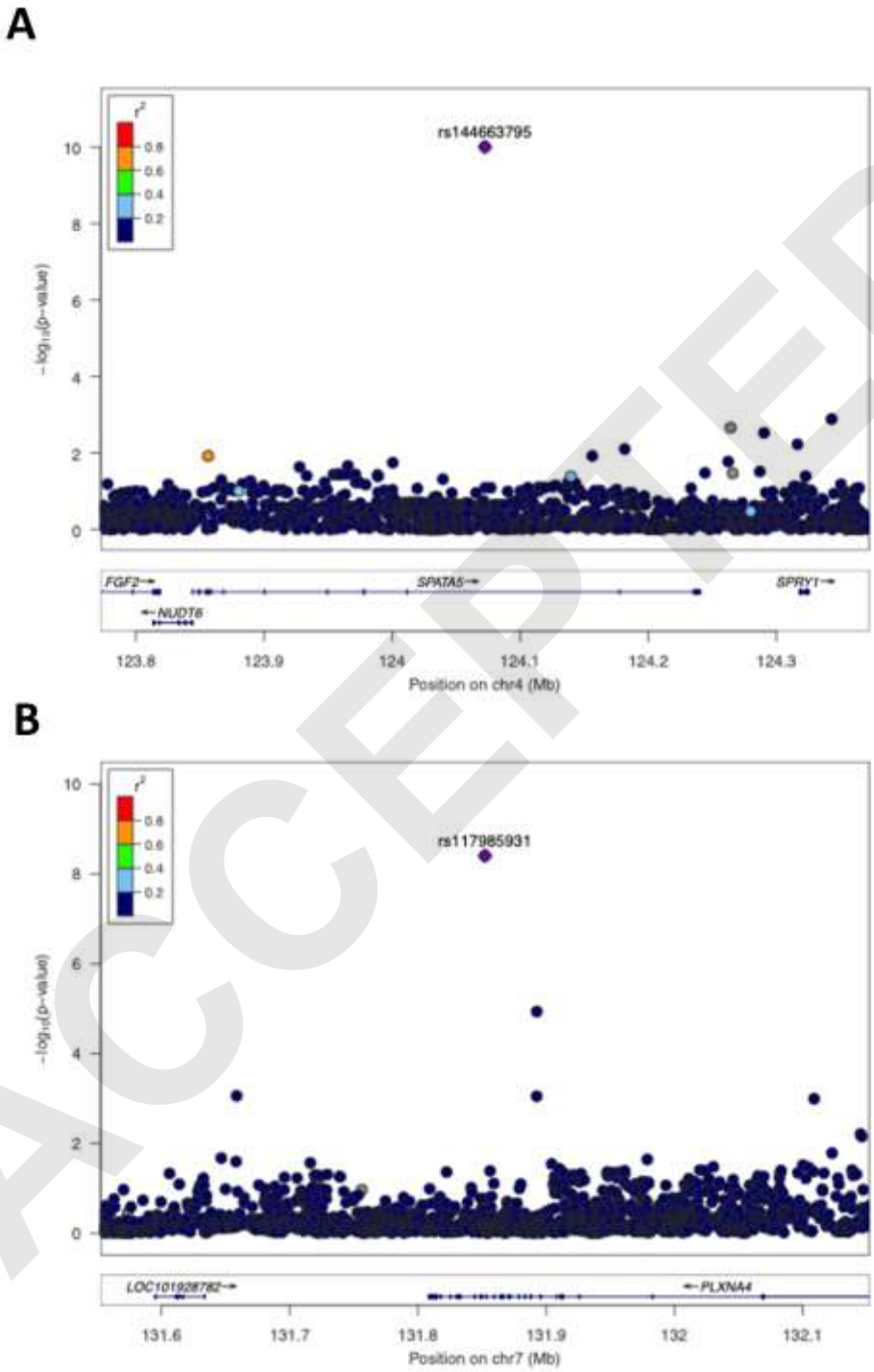

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Table 1. Phenotype Codes

KPRB		number
ICD9-Code	Code Description	
850	Concussion without loss of consciousness	513
850.11	Concussion, with loss of consciousness of 30 minutes or less	373
850.5	Concussion with loss of consciousness	379
850.9	Concussion, unspecified	1227
854.09	Intracranial injury of other and unspecified nature without mention of open intracranial wound, with concussion, unspecified	108
310.2	Traumatic (post-concussional) encephalopathy	1058
ICD10-Code		
F07.81	Postconcussional syndrome	319
S06.0X0A	Concussion without loss of consciousness, initial encounter	286
S06.0X0D	Concussion without loss of consciousness, subsequent encounter	19
S06.0X1A	Concussion with loss of consciousness of 30 minutes or less, initial encounter	90
S06.0X1D	Concussion with loss of consciousness of 30 minutes or less, subsequent encounter	18
S06.0X9A	Concussion with loss of consciousness of unspecified duration, initial encounter	430
S06.0X9D	Concussion with loss of consciousness of unspecified duration, subsequent encounter	139
S06.0X9S	Concussion with loss of consciousness of unspecified duration, sequela	17
S14.0XXA	Concussion and edema of cervical spinal cord, initial encounter	7
	Total unique cases KPRB	3,170
	Total controls KPRB	80,244

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Table 2. Study Demographics

KPRB

	Case	Control	P	risk (%)
Male	1652	41879		3.8
Female	2536	57523	0.0006*	4.2
Height (in)	65.65 (+/- 9.43)	65.73 (+/- 10.17)	0.6745	
weight (lbs)	168 (+/- 37)	172 (+/- 39)	<1E-4	

UK Biobank

	Case	Control	P	risk (%)
Male	479	95992		0.50
Female	416	115612	<1E-4*	0.36
Height (in)	66.6 (6.6)	66.3 (3.6)	0.0067	
weight (lb)	176 (+/- 37)	172 (+/- 35)	0.0009	
age of injury	40.2 (+/- 20.6)	56.7 (+/- 8.0)	<1E-4	

SD in parentheses

*p value vs male

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Table 3. Concussion GWAS and meta-analysis

SNP	Chr	BP	Gene	EA	MAF_KPRB	MAF_UKB
rs144663795	4	124072484	SPATA5	C	0.0022	0.014
rs117985931	7	131852272	PLXNA4	G	0.0013	0.0013

KPRB

SNP	OR_RPGEH (95% CI)	P-value_KPRB
rs144663795	2.65 (1.86 to 3.78)	1.39E-07
rs117985931	1.71 (0.99 to 2.92)	5.79E-02

UK Biobank

SNP	OR_UKB (95% CI)	P_value_UKB
rs144663795	4.77 (2.11 to 10.74)	3.21E-05
rs117985931	11.55 (5.8 to 22.9)	1.10E-14

meta-analysis

SNP	OR_meta	P_fixed	Q	I^2 (95% CI)
rs144663795	2.91	9.74E-11	0.46	0 (0 to 0.90)
rs117985931	3.59	3.97E-09	0	.89 (.15 to .99)

Chr, chromosome
BP, base-pair; Coordinates use Genome Reference Consortium Human Build 37
EA, Effect Allele
MAF, minor allele frequency
OR, odds ratio
Q, Cochran's Q

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Table 4. Validation of previous genetic associations with concussion.

		prior work					this work		
Gene	Allele	Cases	p-value	OR	Phenotype	Ref	Cases	p-value*	OR (95% CI)
ApoE	E2 v E3						3515	0.067	1.12 (.99 to 1.53)
ApoE	E3 v E4	58	0.99	ND	concussion history	15	3515	0.19	0.90 (.76 to 1.06)
ApoE	E3 v E4	165	0.087	1.5	concussion history	11	3515	0.19	0.90 (.76 to 1.06)
ApoE	E3 v E4	133	0.04	0.61	concussion history	17	3515	0.19	0.90 (.76 to 1.06)
ApoE	E3 v E4	28	0.68	ND	concussion history	16	3515	0.19	0.90 (.76 to 1.06)
ApoE	E3 v E4	45	0.015	5.25	total symptom score	13	3515	0.19	0.90 (.76 to 1.06)
ApoE	E3 v E4	30	0.04	ND	chronic brain injury score	12	3515	0.19	0.90 (.76 to 1.06)
ApoE	rs405509	58	0.16	ND	concussion history	15	3515	0.31	1.02 (0.87 to 1.08)
ApoE	rs405510	133	0.31	ND	concussion history	17	3515	0.31	1.02 (0.87 to 1.08)
ApoE	rs405511	48	0.04	8.4	concussion history	44	3515	0.31	1.02 (0.87 to 1.08)
ApoE	rs405509	97	0.03	2.7	concussion history	17	1718	0.30	0.95 (0.86 to 1.04)
BDNF	r6265	165	0.049	2.48	concussion history	11	3515	0.14	0.95 (.80 to 1.11)
DRD4	rs1800955	163	0.014	ND	concussion history	18	1148	0.99	1.00 (0.84 to 0.119)
SLC17A7	rs74174284		0.018	ND	concussion duration and severity	20	3515	0.81	0.99 (0.83 to 1.15)
NGF	rs3783988	196	0.01	ND	better functional outcomes	19	3515	0.63	1.02 (0.87 to 1.17)

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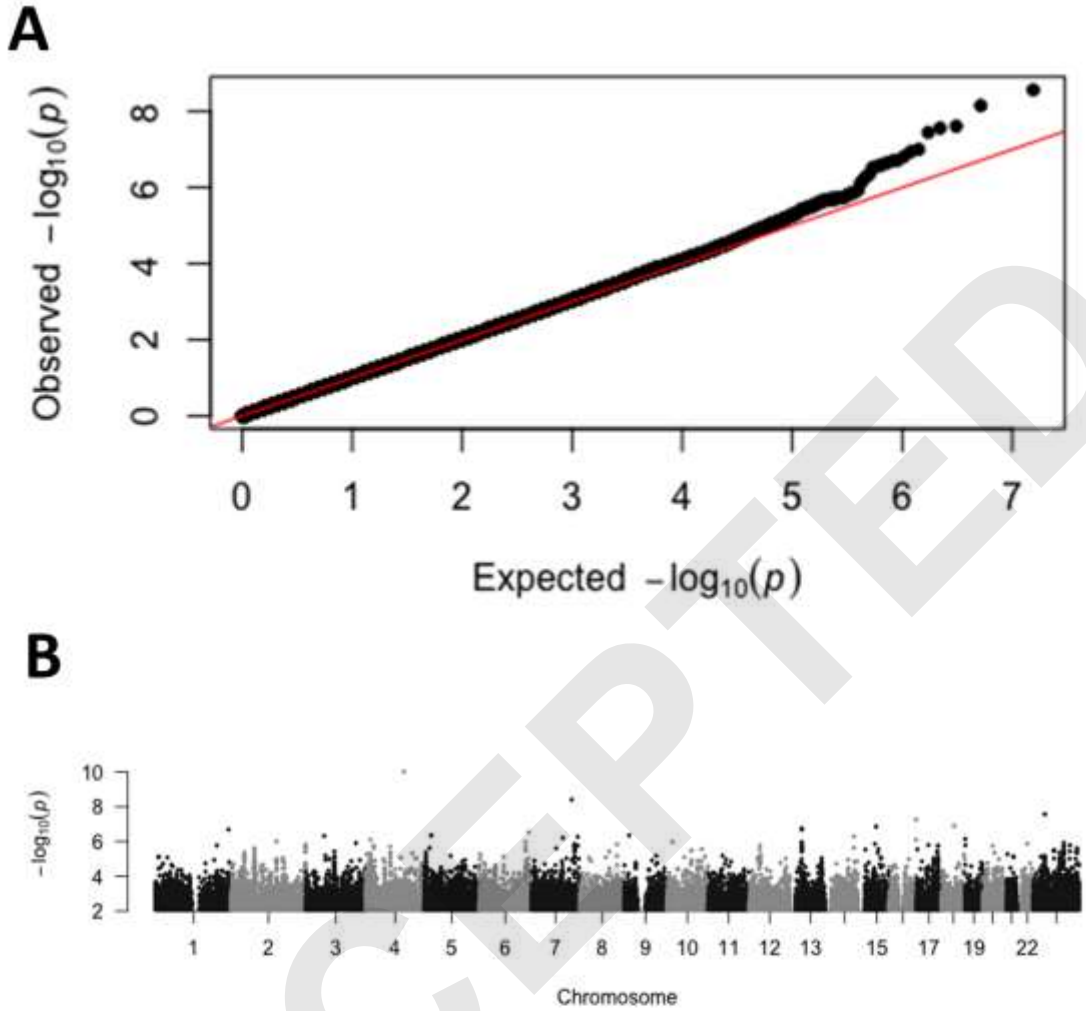

Supplemental Figure 1. A. Quantile-quantile plot for genome-wide association analysis of concussion. The expected versus observed log transformed values for 7,743,986 p-values from the GWAS are graphed. The y-axis shows the observed p-values and the x-axis shows the p-values expected by chance. The black dots represent the SNPs arranged by their observed p-values and the red line shows the expected trajectory if the SNPs had p-values expected by chance. B. Manhattan plot for genome-wide association analysis of concussion. The -log10 p-values for association with concussions for SNPs from the GWAS are plotted by genomic position with chromosome number listed across the bottom. The y-axis shows the -log10 p-value for association with concussion.

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